TalkBox

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Introduction

Background

Non-verbal individuals often use speech playback devices to communicate. Dr. Foad Hamidi of the UMBC Department of Information Systems has been developing an open-source "do-it-yourself" playback device called the TalkBox that can be assembled and customized by the user to enable introductory-level communication for users to interact with others.

Objective

Improve upon the existing prototypes of the TalkBox to make the device:

1. Modular
2. More durable
3. More aesthetically pleasing

Key System Requirements

- The system shall be adaptable to any arrangement of user inputs with a minimum of 2 inputs to a maximum of 12 inputs.
- The system shall be modular so that the user may change the amount of inputs and the message projected by each input by changing the input mechanism and audio files.
- Connections between the input and output shall be designed for at least 100,000 cycles of connecting and disconnecting.
- Input devices shall require no more than 1 N to activate.
- Each input position shall correspond to a specific audio output.

Functional Diagrams

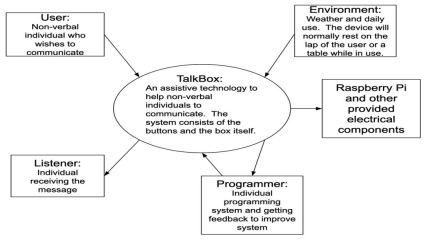

Figure 1. The System Boundary Diagram for the TalkBox. This details the interfaces between the physical boundaries of the TalkBox and other variables and systems.

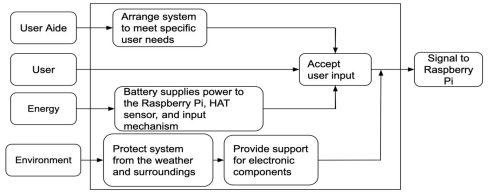

Figure 2. The Functional Block Diagram for the TalkBox. This demonstrates the functions that occur within the physical boundaries of the system, culminating with the transmission of a signal to the Raspberry Pi.

Prototypes

First Prototype

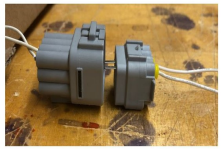

Figure 3 and Figure 4. The objective of the first prototype was to establish a secure electrical connection. This connection was confirmed by an observed voltage drop between the Raspberry Pi and the end of the wire.

Second Prototype

Figure 5. The objective of the second prototype was to create a more robust input device which could accept multiple inputs. A new electrical enclosure was also constructed to determine the orientation of the internal components.

Final Prototype

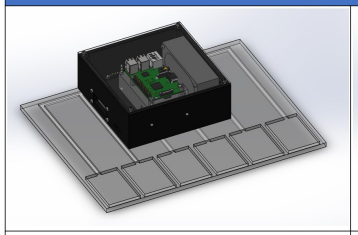

Figure 6. The TalkBox prototype assembled on SolidWorks. This details the finalized design for a TalkBox which meets the previously established requirements.

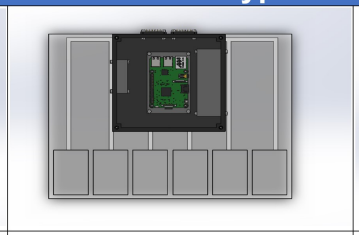

Figure 7. A plan view of the final prototype design created on SolidWorks. Shows combination of the electrical enclosure, electrical fixtures, and input device.

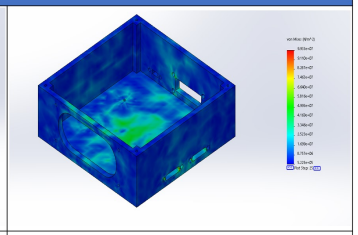

Figure 8. Finite Element Analysis for a 1-m drop test to show the stress distribution across the surface of the electrical enclosure.

Observations and Conclusions

- By using electrical connections we were able to increase the modularity of the TalkBox by separating the electrical components
- Wires and the rigid fixtures used to secure each electrical component further increased the necessary volume of the enclosure.
- Laser cutting was a more suitable method of rapid prototyping for two-dimensional parts than three-dimensional parts.
- 3D printed parts were subject to warping and the 3D printers themselves were subject to clogging, which increased the lead time for some of the prototypes.

Figure 9. During production of the third prototype, the bottom layer of the box was warped, presumably due to a printing error.

Future Work

- Finish construction of the final prototype of the TalkBox and test robustness of new design compared to prior iterations.
- Purchase shorter cables or seek alternative methods of cable management so that the volume of the enclosure can be decreased.

Acknowledgements

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